AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-1230	**Classification:**	Major / Water of Hydration Content (CQA)
Product / API Name:	Ciprofloxacin Monohydrochloride Monohydrate API	**Batch / Lot Number:**	CIP-2026-304
Manufacturing Stage:	Stage 4: Centrifugal Isolation & Atmospheric Drying	**Equipment Tag:**	Peeler Centrifuge CF-03 / Vacuum Dryer VCD-01
Date of Occurrence:	2026-10-16 07:15 UTC	**Date of Discovery:**	2026-10-17 13:00 UTC
Material / Lot ID:	Wash Water Purified WTR-PW-449	**Reporting Unit:**	QC Chemical Analysis Laboratory

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Water Content by Karl Fischer	**4.7% - 6.7% (Monohydrate state)**	**7.8% (w/w) [OOS]**	Coulometric Karl Fischer Titration
Chromatographic Purity (HPLC)	Total Impurities NMT 0.50%	0.18%	Ph. Eur. HPLC Method
Assay (Titrimetric)	98.0% - 102.0%	99.3%	Non-aqueous titration with HClO4
Residue on Ignition	NMT 0.10%	0.04%	Sulfated Ash 600°C

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

Karl Fischer titration for Ciprofloxacin HCl Monohydrate batch CIP-2026-304 yielded water content of 7.8%, exceeding the monohydrate theoretical boundary of 6.7%. Investigation of peeler centrifuge CF-03 wash manifold demonstrated that cake wash spray nozzle #2 had clogged, causing uneven displacement of mother liquor and leaving excessive surface water and hygroscopic mother liquor trapped within the lower centrifuge cake layer.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. 1,800 kg of Ciprofloxacin HCl quarantined in sealed polyethylene fiber drums.
2. Centrifuge CF-03 wash spray header dismounted for ultrasonic cleaning and spray pattern validation.
3. Fluidized bed finish drying re-pass authorized under Deviation Protocol DEV-1230-DP.

Prepared by:	F. Moreau (QC Analyst)	2026-10-17
Verified by:	J. Brennan (Production Manager)	2026-10-17
QA Approval:	Dr. S. Thornton (QA Lead)	**Status: PENDING ROOT CAUSE ANALYSIS**